How-To Library

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Enabling Kerberos For Hive and Impala Databases in Dynamic Data Masking

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Abstract

Kerberos is an effective authentication protocol built on symmetric-key cryptography. This article describes how to enable Kerberos authentication for Hive and Impala databases in Dynamic Data Masking.

Supported Versions

- Dynamic Data Masking 9.9.1

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Kerberos for Hive and Impala Databases

Dynamic Data Masking supports the Kerberos authentication protocol for Hive and Impala databases. Kerberos provides strong authentication for client/server applications through secret-key cryptography.

To enable Kerberos authentication, you first configure the Hive or Impala server. In Dynamic Data Masking, configure the `ddm.security` file with Kerberos-specific details, such as the location of the `krb5.conf` file. When you create the Hive or Impala database node, you must include details such as the JDBC URL for the Hive or Impala driver, and the database server principal.

Prerequisites

The DDM services for Impala and Hive use details from the `krb5.conf` file. Before you configure the DDM for Hive or Impala services for Kerberos, you must copy the `krb5.conf` file from the Hadoop server to the Dynamic Data Masking Server. You must also create a service principal name and a keytab for Dynamic Data Masking. It is also important to verify that SSL is enabled for Hive or Impala, and to have the appropriate JDBC drivers for Hive or Impala.

Verify the following prerequisites before you configure Kerberos authentication in Dynamic Data Masking:

1. Create a service principal name, or SPN, for Dynamic Data Masking. The SPN is a unique identifier of a service instance. The Dynamic Data Masking SPN must have the privileges to run a map reduce job. The SPN must also have access to all of the tables and databases in addition to all of the database files in HDFS. The typical format of a SPN is: servicename/fqdn@realm.com
2. You must create a keytab for Dynamic Data Masking that contains the details of the Dynamic Data Masking SPN. A keytab is a file that contains pairs of Kerberos principals and encrypted keys, that are derived from the Kerberos password.
3. Copy the configured `krb5.conf` file from the Hadoop server node to the Dynamic Data Masking Server. The `krb5.conf` file contains details about the Kerberos configuration, including the locations of key distribution center servers (KDCs) and admin servers, and other details.
4. If SSL is enabled for Hive or Impala, copy the server public certificates to the Dynamic Data Masking Server.
5. You must have the appropriate JDBC drivers for the Hive or Impala database.

Connection Management

Select the Hive or Impala database type to add a Hive or Impala database connection node to the Management Console tree.

You can use the Hive or Impala database type to access Hadoop-compatible file systems. To connect to multiple Hive or Impala databases, create database nodes in the Management Console and enter different sets of driver files in the classpath parameter.

Click **Test Connection** to verify that the Dynamic Data Masking service can access the database.

Hive Connection Parameters

Define the following connection parameters for a Hive database:

DDM Database Name

Name for the database that appears in the Management Console tree.

Hive Database Name

Database name for the Hive database.

Driver Classpath

Classpath of the Hive database driver files on the Dynamic Data Masking Server. Use semicolons to separate multiple classpaths.

You can use an asterisk (*) to indicate all the JAR files in a directory. Dynamic Data Masking ignores files in the directory that are not JAR files. For example, you might enter the following location for Windows:

```
C:\JDBC_Drivers\Hive\hive\*;C:\JDBC_Drivers\Hive\hadoop\*
```

Driver Class Name

Fully qualified class name of the Hive driver specified in the **Driver Classpath** property.

Connect String (URL)

JDBC connection URL based on the driver.

If the Hive database has Kerberos authentication enabled, the URL properties **auth** and **kerberosAuthType** are mandatory.

For example: `jdbc:hive2://hiveserver:10000/default;principal=hive/hiveserver@realm.com;` **auth=kerberos;kerberosAuthType=fromSubject**

Database Server Principal

When you connect to a Hive database with Kerberos authentication enabled, specify the server principal of Hive, even though it is included in the JDBC URL. Otherwise, leave this parameter blank.

Key Store

Select Custom if you have configured a custom keystore. Select Default if you want to use the default keystore preconfigured for use with Dynamic Data Masking.

DBA Username

Optional user name for the database user account to log in to the Hive database. The database user must be a privileged user that has SELECT access to all the tables that the client user has SELECT access to. This parameter is valid for the default keystore.

Leave this parameter blank if the Hive database has Kerberos authentication enabled.

DBA Password

Optional password for the database user. This parameter is valid for the default keystore.

Leave this parameter blank if the Hive database has Kerberos authentication enabled.

Key Store Name

Name of the custom keystore, defined in the `ddm.security` file. This parameter is valid for custom keystores.

Alias

Alias name for the custom keystore. For CyberArk accounts, the alias name was defined during creation of the CyberArk account. This parameter is valid for custom keystores.

Impala Connection Parameters

Define the following connection parameters for an Impala database:

DDM Database Name

Name for the database that appears in the Management Console tree.

Impala Database Name

Database name for the Impala database.

Driver Classpath

Classpath of the Impala database driver files on the Dynamic Data Masking Server. Use semicolons to separate multiple classpaths.

You can use an asterisk (*) to indicate all the JAR files in a directory. Dynamic Data Masking ignores files in the directory that are not JAR files.

Example for Windows:

```
C:\JDBC_Drivers\impala\*
```

Example for Linux:

```
/root/cloudera/impala/*
```

Driver Class Name

Fully qualified class name of the Hive driver specified in the **Driver Classpath** property.

Example:

```
org.apache.hive.jdbc.HiveDriver
```

Connect String (URL)

JDBC connection URL based on the driver.

If the Impala database has Kerberos authentication enabled, and the Hive driver is used to connect to Impala, the URL properties **auth** and **kerberosAuthType** are mandatory.

Example: `jdbc:hive2://impalaserver:21050/default;principal=impala/impalaserver@realm.com;auth=kerberos;kerberosAuthType=fromSubject`

Database Server Principal

When you connect to an Impala database with Kerberos authentication enabled, specify the server principal of Impala, even though it is included in the JDBC URL. Otherwise, leave this parameter blank.

Key Store

Select Custom if you have configured a custom keystore. Select Default if you want to use the default keystore preconfigured for use with Dynamic Data Masking.

DBA Username

Optional user name for the database user account to log in to the Impala database. The database user must have SELECT access privileges for all the tables to which the client user has SELECT access privileges. This parameter is valid for the default keystore.

Leave this parameter blank if the Impala database has Kerberos authentication enabled.

DBA Password

Optional password for the database user. This parameter is valid for the default keystore.

Leave this parameter blank if the Impala database has Kerberos authentication enabled.

Key Store Name

Name of the custom keystore, defined in the `ddm.security` file. This parameter is valid for custom keystores.

Alias

Alias name for the custom keystore. For CyberArk accounts, the alias name was defined during creation of the CyberArk account. This parameter is valid for custom keystores.

Server Configuration

When a client connects to the database through an intermediary service like Dynamic Data Masking, the database has a connection associated with the intermediary service user account instead of the database client user account. You can change the user context through a database-specific property in the JDBC/ODBC properties, regardless of the user making the connection. Dynamic Data Masking adds this property when it connects to a Hive or Impala database.

To configure a Hive server for user context delegation, enable impersonation of all users and provide proxy access to the Dynamic Data Masking SPN. To configure an Impala server, enable the Dynamic Data Masking SPN as a proxy user.

The server configuration steps might not be the same for all distributions of Apache Hive. The configuration information provided below is for the Cloudera and Hortonworks distributions of Hive and Impala.

Hive Server Configuration

For Dynamic Data Masking to work with Hive, you must enable impersonation on the HiveServer2 and provide proxy access to the Dynamic Data Masking SPN. You define the Dynamic Data Masking service principal in the `ddm.security` file.

The following table lists HiveServer2 configuration properties:

Database	Property	Description
Cloudera Hive2	HiveServer2 Enable Impersonation	Enables impersonation. In the Cloudera Manager Hive2 service configuration, select the **HiveServer2DefaultGroup** check box. This is not required if Sentry is configured.
Cloudera Hive2	hadoop.proxyuser.<ProxyUser>.hosts	Enables proxy access to the Dynamic Data Masking Server principal. Edit the Hive Service Advanced Configuration Snippet (Safety Valve) for `core-site.xml` to set the property value to " * ". The * value enables impersonation for all users. When you configure the property, replace <ProxyUser> with your user name in the Dynamic Data Masking principal. For example, if the Dynamic Data Masking service principal is ddmserver/ddmhost@realm.com: hadoop.proxyuser.**ddmserver**.hosts value = *
Cloudera Hive2	hadoop.proxyuser.<ProxyUser>.groups	Edit the Hive Service Advanced Configuration Snippet (Safety Valve) for `core-site.xml` to set the property value to " * ". The * value enables impersonation for all users. When you configure the property, replace <ProxyUser> with your user name in the Dynamic Data Masking principal. For example, if the Dynamic Data Masking service principal is ddmserver/ddmhost@realm.com: hadoop.proxyuser.**ddmserver**.groups value = *

Database	Property	Description
Hortonworks	hadoop.proxyuser.<ProxyUser>.hosts	Edit `core-site.xml` to set both property values to " * ". The * value enables impersonation for all users. When you configure the properties, replace <ProxyUser> with your user name in the Dynamic Data Masking principal. For example, if the Dynamic Data Masking service principal is ddmserver/ddmhost@realm.com: hadoop.proxyuser.**ddmserver**.hosts value = * hadoop.proxyuser.**ddmserver**.groups value = *
Hortonworks	hadoop.proxyuser.<ProxyUser>.groups	Edit `core-site.xml` to set both property values to " * ". The * value enables impersonation for all users. When you configure the properties, replace <ProxyUser> with your user name in the Dynamic Data Masking principal. For example, if the Dynamic Data Masking service principal is ddmserver/ddmhost@realm.com: hadoop.proxyuser.**ddmserver**.hosts value = * hadoop.proxyuser.**ddmserver**.groups value = *

Impala Server Configuration

For Dynamic Data Masking to work with a Cloudera distribution of Impala, enable the Dynamic Data Masking SPN as a proxy user in the Impala service.

Edit the Impala Daemon Command Line Argument Advanced Configuration Snippet (Safety Valve) to set the proxy user. For example: `authorized_proxy_user_config=<ProxyUser>=*`

When you configure the property, replace <ProxyUser> with the Dynamic Data Masking principal. For example, if the Dynamic Data Masking service principal is ddmserver/ddmhost@realm.com:

`authorized_proxy_user_config=`**`ddmserver`**`=*`

You define the Dynamic Data Masking service principal in the `ddm.security` file.

Enabling Kerberos Authentication in Dynamic Data Masking

To enable Kerberos authentication in Dynamic Data Masking, copy the `krb5.conf` file and the keytab file for the Dynamic Data Masking service principal to the Dynamic Data Masking Server machine. Then configure the `ddm.security` file.

1. Stop the Dynamic Data Masking Server.
2. Copy the `krb5.conf` file and the keytab file for the Dynamic Data Masking service principal to the Dynamic Data Masking Server machine.
3. If you have not created an XML `ddm.security` configuration file, create the file in the following location: `<DDM>/cfg/ddm.security`

4. Configure the `ddm.security` file as shown in the following example:

```
<XML>
 <kdc>/etc/krb5.conf</kdc>
 <jaasConfig type="ArrayList">
    <entry type="JaasDescriptor">
        <fqcn>com.activebase.security.jaas.JaasProcessorImpl</fqcn>
        <configuration>
            <jaasEntries type="HashMap">
                <entry>
                    <value type="ArrayList">
                        <entry type="HashMap">
                            <entry>
                                <value>com.sun.security.auth.module.Krb5LoginModule</value>
                                <key>moduleClass</key>
                            </entry>
                            <entry>
                                <value>required</value>
                                <key>moduleFlag</key>
                            </entry>
                            <entry>
                                <value type="HashMap">
                                    <entry>
                                        <value>ddmserver/ddmhost@realm.com</value>
                                        <key>principal</key>
                                    </entry>
                                    <entry>
                                        <value>cfg/ddmService.keytab</value>
                                        <key>keyTab</key>
                                    </entry>
                                </value>
                                <key>options</key>
                            </entry>
                        </entry>
                    </value>
                    <key>default</key>
                </entry>
            </jaasEntries>
        </configuration>
    </entry>
</jaasConfig>
</XML>
```

5. Configure the values in the `ddm.security` file above.
    - For the value of the "principal" key, give the Dynamic Data Masking SPN.
    - For the value of the "keyTab" key, give the keytab file name with path.

If SSL authentication is enabled on the Hive or Impala Server, follow the steps in Chapter 3, topics "Keystore Configuration" and "Truststore Configuration" to enable SSL for Hive or Impala services.

JDBC Client Configuration

Configure the JDBC clients for use with Kerberos authentication.

The following table describes typical JDBC client configuration changes required:

JDBC URL (Direct)	JDBC URL (Dynamic Data Masking)	Description
jdbc:hive2://<HiveServer2 Host>:<HiveServer2 Port>/;principal=<Server Principal of Hive >;SaslQOP=auth;auth=kerberos	jdbc:hive2://<DDM Server Host>:<DDM Server Port>/;principal=<Server Principal of DDM>;SaslQOP=auth;auth=kerberos	The Dynamic Data Masking host and port in the client JDBC URL must point to the Dynamic Data Masking Hive or Impala service host and port. The principal in the client JDBC URL must be the Dynamic Data Masking Server principal name.

The following table describes typical JDBC client configuration changes required when SSL authentication is enabled on Hive or Impala:

JDBC URL (Direct)	JDBC URL (Dynamic Data Masking)	Description
jdbc:hive2://<HiveServer2 Host>:<HiveServer2 Port>/;principal=<Server Principal of Hive>; SaslQOP=auth;auth=kerberos;ssl=true;sslTrustStore=<path of truststore that contains the Hive certificate>;trustStorePassword=<truststore password>	jdbc:hive2://<DDM Server Host>:<DDM Server Port>/;principal=<Server Principal of DDM>;SaslQOP=auth;auth=Kerberos;ssl=true;sslTrustStore=<path of truststore that contains the DDM certificate>;trustStorePassword=<truststore password>	The Dynamic Data Masking host and port in the client JDBC URL must point to the Dynamic Data Masking Hive or Impala service host and port. The principal in the client JDBC URL must be the Dynamic Data Masking Server principal name.

Configuring the Hue Service

Use the Hue Service Advanced Configuration Snippet (Safety Valve) for `hue_safety_valve.ini` and `hive-site.xml` to configure the Hue service to make connections through Dynamic Data Masking.

1. Configure the following properties in the Hue Service Advanced Configuration Snippet (Safety Valve) for `hue_safety_valve.ini`:

   ```
   [beeswax]
   hive_server_host=DDMHost
   hive_server_port=DDMPort

   [impala]
   server_host=DDMHost
   server_port=DDMPort
   impala_principal=<server principal of Dynamic Data Masking>
   ```

   If SSL authentication is enabled on Hive or Impala, configure the following properties:

   ```
   [beeswax]
   hive_server_host=DDMHost
   hive_server_port=DDMPort
   [[ssl]]
   cacerts=<path of public certificate of Dynamic Data Masking>

   [impala]
   server_host=DDMHost
   server_port=DDMPort
   impala_principal=<server principal of Dynamic Data Masking>
   [[ssl]]
   cacerts=<path of public certificate of Dynamic Data Masking>
   ```

2. Add the following property in the Hue Service Advanced Configuration Snippet (Safety Valve) for `hive-site.xml`:

   hive.server2.authentication.kerberos.principal
   Give the value of the property as the Dynamic Data Masking SPN. For example: `ddmserver/ddmhost@realm.com`

JAAS Configuration Options in the ddm.security File

When you configure the `cfg/ddm.security` for Kerberos authentication, SSL authentication, or for use with a custom security provider, you use Java Authentication and Authorization Service configuration options.

The following table describes the JAAS parameters supported in the `cfg/ddm.security` file:

Option	Description
KDC	The path to the configuration file that provides details of the Kerberos key distribution center.
jaasConfig	The start of JAAS configuration.
JaasDescriptor	The JAAS descriptor that provides JAAS processor and configuration entries for the processor.
fqcn	The full-qualified class name of the JAAS processor implementation. Dynamic Data Masking provides one implementation: com.activebase.security.jaas.JaasProcessorImpl
jaasEntries	The map of JAAS configuration entries.
entry	The JAAS configuration entry.
key	The name of the configuration entry. Dynamic Data Masking supports one mandatory entry called "default."
value	The list that contains the map of login modules and their configuration parameters.
entry	The configuration entry of the login module.
moduleClass	The login module class tag.
value (moduleClass)	The fully-qualified class name of the login module implementation. Dynamic Data Masking supports two implementations: 1. com.sun.security.auth.module.Krb5LoginModule 2. com.ibm.security.auth.module.Krb5LoginModule
moduleFlag	The login module flag tag.

Option	Description
value (moduleFlag)	Java supports the following standard options: REQUIRED, REQUISITE, SUFFICIENT, and OPTIONAL.
options	The configuration options of the login module as a map of the *option name* and the corresponding*value*. Dynamic Data Masking supports the following configuration options of the com.sun.security.auth.module.Krb5LoginModule login module: - principal - keyTab - useKeyTab - storeKey - doNotPrompt - isInitiator - useTicketCache - refreshKrb5Config - renewTGT - storePass - clearPass - useFirstPass - debug For more information on these options, refer to the following Oracle documentation: https://docs.oracle.com/javase/8/docs/jre/api/security/jaas/spec/com/sun/security/auth/module/Krb5LoginModule.html Dynamic Data Masking supports the following configuration options of the com.ibm.security.auth.module.Krb5LoginModule login module: - KRB5CCNAME - principal - UseDefaultCcache - ticketcache - credsType - both For more information on these options, refer to the following IBM documentation: https://www.ibm.com/support/knowledgecenter/en/SSYKE2_6.0.0/com.ibm.java.security.component.60.doc/security-component/jgssDocs/jaas_login_user.html

Sample ddm.security File

The following image is an example of the `cfg/ddm.security` file configured with the JAAS entries listed in the table above:

```
<kdc>/etc/krb5.conf</kdc>

<jaasConfig type="ArrayList">
    <entry type="JaasDescriptor">
        <fqcn>com.activebase.security.jaas.JaasProcessorImpl</fqcn>
        <configuration>
            <jaasEntries type="HashMap">
                <entry>
                    <key>default</key>
                    <value type="ArrayList">
                        <entry type="HashMap">
                            <entry>
                                <key>moduleClass</key>
                                <value>com.sun.security.auth.module.Krb5LoginModule</value>
                            </entry>
                            <entry>
                                <key>moduleFlag</key>
                                <value>required</value>
                            </entry>
                            <entry>
                                <key>options</key>
                                <value type="HashMap">
                                    <entry>
                                        <key>principal</key>
                                        <value>ddm/inkr74dsg695.informatica.com@KERB</value>
                                    </entry>
                                    <entry>
                                        <key>keyTab</key>
                                        <value>/root/ddm.keytab</value>
                                    </entry>
                                    <entry>
                                        <key>doNotPrompt</key>
                                        <value>true</value>
                                    </entry>
                                    <entry>
                                        <key>useKeyTab</key>
                                        <value>true</value>
                                    </entry>
                                </value>
                            </entry>
                        </entry>
                    </value>
                </entry>
            </jaasEntries>
        </configuration>
    </entry>
```

Enabling Kerberos Encryption in Dynamic Data Masking

You can configure Kerberos encryption by setting the `sasl.qop` value to `auth-conf` for hive databases. Copy the `krb5.conf` file and the keytab file for the Dynamic Data Masking service principal to the Dynamic Data Masking Server machine. Then configure the `ddm.security` file.

1. Stop the Dynamic Data Masking Server.
2. On Windows, remove the Dynamic Data Masking Service.
3. Copy the `krb5.conf` file and the keytab file for the Dynamic Data Masking service principal to the Dynamic Data Masking Server machine.
4. If you have not already created an XML `ddm.security` configuration file, create the file in the following location: `<DDM>/cfg/ddm.security`
5. Configure the `ddm.security` configuration file. For more information, see "Configuring the ddm.security file."
6. Start the Dynamic Data Masking Server.

Note: Add the port that you configured in the `ddm.security` file to the service. In the example, the `ddm.security` file is configured for the Dynamic Data Masking for Hive service running on port 10001. In this case, you would add port 10081 to the Dynamic Data Masking for Hive service.

7. Configure the JDBC or ODBC client. For more information, see "Client Configuration."

Configuring the ddm.security file

To configure Kerberos encryption for Hive databases, you must specify the following values in the `ddm.security` file.

1. For the value of the principal key, specify the Dynamic Data Masking SPN.
   For example: `ddmserver/ddmhost@realm.com`
2. For the value of the keyTab key, specify the keytab file name with path.
   For example: `cfg/ddmService.keytab`
3. If Hive is configured with auth-conf, define the port strategy for the service:
   - Key is the port number of Dynamic Data Masking for the Hive service.
   - Value is a map with the key as `sasl.qop` and the value as `auth-conf`.

   For example, the following `ddm.security` file is configured for the Dynamic Data Masking for Hive service running on port 10001:

```
<XML>
 <kdc>/etc/krb5.conf</kdc>
 <jaasConfig type="ArrayList">
    <entry type="JaasDescriptor">
        <fqcn>com.activebase.security.jaas.JaasProcessorImpl</fqcn>
        <configuration>
            <jaasEntries type="HashMap">
                <entry>
                    <value type="ArrayList">
                        <entry type="HashMap">
                            <entry>
                                <value>com.sun.security.auth.module.Krb5LoginModule</value>
                                <key>moduleClass</key>
                            </entry>
                            <entry>
                                <value>required</value>
                                <key>moduleFlag</key>
                            </entry>
                            <entry>
                                <value type="HashMap">
                                    <entry>
                                        <value>ddmserver/ddmhost@realm.com</value>
                                        <key>principal</key>
                                    </entry>
                                    <entry>
                                        <value>cfg/ddmService.keytab</value>
                                        <key>keyTab</key>
                                    </entry>
                                </value>
                                <key>options</key>
                            </entry>
                        </entry>
                    </value>
                    <key>default</key>
                </entry>
            </jaasEntries>
        </configuration>
    </entry>
</jaasConfig>
<serviceStrategies type="ArrayList">
    <entry type="StrategyDescriptor">
        <fqcn>com.activebase.security.service.strategies.PortStrategy</fqcn>
        <configuration>
            <ports type="HashMap">
                <entry>
                    <key>10001</key>
                    <value type="HashMap">
                        <entry>
                            <key>sasl.qop</key>
                            <value>auth-conf</value>
                        </entry>
                    </value>
                </entry>
```

```
            </ports>
        </configuration>
    </entry>
</serviceStrategies>
</XML>
```

Client Configuration

After you configure the `ddm.security` file, configure the client.

JDBC Client Configuration

The Dynamic Data Masking host and port in the client JDBC URL must point to the Dynamic Data Masking Hive service host and port. The principal in the client JDBC URL must be the Dynamic Data Masking Server principal name.

You can connect to the client using a direct connection or you can connect to the client through Dynamic Data Masking. In the client connection URL, add the property `SaslQOP` using one of the following formats:

JDBC URL (Direct Connection)

If you are connecting to the client directly, use the following format:

```
jdbc:hive2://<HiveServer2 Host>:<HiveServer2 Port>/;principal=<Server Principal of Hive
>;SaslQOP=auth-conf;
```

JDBC URL (Connection through Dynamic Data Masking)

If you are connecting to the client through Dynamic Data Masking, use the following format:

```
jdbc:hive2://<DDM Server Host>:<DDM Service Port configured in
ddm.secuirty>/;principal=<Server Principal of DDM>;SaslQOP=auth-conf;
```

ODBC Client Configuration

Configure the ODBC client through the Windows ODBC Data Source Administrator and create the ODBC data source. When you create the ODBC Data Source, select **Thrift Transport** > **SASL**.

Authors

Jennifer Osburn

Julie Henry

Acknowledgements

Thanks to the Dynamic Data Masking development team for their help in completing this article.